

# Journal of Graduate Medical Education

## Generative Artificial Intelligence Use in Pediatric Residency Personal Statements

--Manuscript Draft--

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| Abstract:          | <p>Background</p> <p>Use of artificial intelligence (AI, i.e. ChatGPT) has caused concern about personal statement authenticity in residency applications. This may threaten holistic review and ability to assess applicant characteristics. Humans struggle to determine which personal statements are AI versus human-generated.</p> <p>Objective</p> <p>To assess the frequency of AI use and percent of personal statement text generated by AI among pediatric residency applications using 2 AI detection platforms.</p> <p>Methods</p> <p>Retrospective cross-sectional study of de-identified residency applications from a single pediatric program between 2022-2024. Demographic information and interview status were abstracted. Personal statements were entered into two AI detectors (Quillbot, ZeroGPT) which determined AI% generated text. Outcomes were compared across applicant characteristics and time.</p> <p>Results</p> <p>3521 applications were assessed (1255 in 2022, 1139 in 2023, 1157 in 2024). The correlation between the AI% detected in each platform was .68 (95% CI .66, .7). Mean AI% detection across all years was 11.7% (SD 26.3) in Quillbot and 13% (SD 20.5) in ZeroGPT. A significant increase in mean AI% was detected in both platforms between 2022 and 2024 (<math>p&lt;.001</math>). Significant differences in AI% were found by self-identity and medical school country (<math>p&lt;.001</math>). In adjusted analysis, no statistically significant difference existed in AI% detection among applicants invited to interview and those who were not invited (mean difference .5, 95% CI -1.8, 2.8).</p> <p>Conclusion</p> <p>AI use in pediatric residency personal statements is increasing. Significant variability in AI detection exists across platforms, questioning their accuracy. Stakeholders should consider how to integrate and monitor AI use among applicants.</p> |

## Generative Artificial Intelligence Use in Pediatric Residency Personal Statements

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## Abstract

**Background:** Use of artificial intelligence (AI, i.e. ChatGPT) has caused concern about personal statement authenticity in residency applications. This may threaten holistic review and ability to assess applicant characteristics. Humans struggle to determine which personal statements are AI versus human-generated. **Objective:** To assess the frequency of AI use and percent of personal statement text generated by AI among pediatric residency applications using 2 AI detection platforms. **Methods:** Retrospective cross-sectional study of de-identified residency applications from a single pediatric program between 2022-2024. Demographic information and interview status were abstracted. Personal statements were entered into two AI detectors (Quillbot, ZeroGPT) which determined AI% generated text. Outcomes were compared across applicant characteristics and time. **Results:** 3521 applications were assessed (1255 in 2022, 1139 in 2023, 1157 in 2024). The correlation between the AI% detected in each platform was .68 (95% CI .66, .7). Mean AI% detection across all years was 11.7% (SD 26.3) in Quillbot and 13% (SD 20.5) in ZeroGPT. A significant increase in mean AI% was detected in both platforms between 2022 and 2024 ( $p<.001$ ). Significant differences in AI% were found by self-identity and medical school country ( $p<.001$ ). In adjusted analysis, no statistically significant difference existed in AI% detection among applicants invited to interview and those who were not invited (mean difference .5, 95% CI -1.8, 2.8). **Conclusion:** AI use in pediatric residency personal statements is increasing. Significant variability in AI detection exists across platforms, questioning their accuracy. Stakeholders should consider how to integrate and monitor AI use among applicants.

## Introduction:

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4 The emerging use of generative artificial intelligence (AI) by the general public is raising  
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6 concerns about personal statement authenticity in residency applications. Personal statements  
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8 have been a longstanding and integral part of residency applications, allowing applicants to share  
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10 and expand on their innate personal characteristics and express their “goodness of fit” for a  
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12 particular specialty or program.<sup>1-3</sup> Prior to AI, plagiarism and fabrication in personal statements  
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14 were already of concern, and the increasingly widespread use of AI may contribute further to  
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16 concerns regarding authenticity. Currently, the Electronic Residency Application System  
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18 (ERAS) advises: “The use of AI tools is acceptable for brainstorming, proofreading, or editing  
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20 the personal statement, but the final submission should represent your own work.”<sup>5</sup>

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26 Despite program directors’ (PDs) debate over the value of personal statements in the  
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28 screening process for residency interviews, many PDs rank them highly in terms of importance.<sup>6-</sup>  
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31 <sup>10</sup> Historically, residency PDs have relied heavily on United States Medical Licensing Exam  
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33 (USMLE) scores when screening residency applications to make interview decisions.<sup>7,11</sup> Yet,  
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35 with increasing numbers of medical schools transitioning from traditional grading schemes to  
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37 pass/fail evaluations, and the USMLE Step 1 becoming pass/fail, it has become increasingly  
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39 difficult for residency PDs to rely on these objective measures as part of their applicant screening  
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41 process and is placing more emphasis on other applicant characteristics, such as personal  
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43 statements. PDs who weigh personal statements highly in the recruitment process believe they  
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45 offer meaningful insight into applicants’ communication skills, allow applicants to highlight  
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47 unique attributes, and articulate how their values align with a program’s mission and vision.<sup>2,8,12</sup>  
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52 PDs agree that personal statements are more meaningful when they share insight into the  
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54 decision to pursue a specific specialty training, or elaborate upon their vision for potential career  
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56 plans after training.<sup>2,12</sup>  
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4 Long before the advent of AI tools, some applicants used paid services to strengthen their  
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6 residency applications, such as hiring contractors to review or revise personal statements.<sup>13</sup> The  
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8 emergence of publicly available generative AI services (i.e. ChatGPT) represents a natural  
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10 extension of this longstanding practice and offers a lower-cost and more accessible alternative to  
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12 traditional editing services. Since the public release of ChatGPT-4 in 2022, there has been an  
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14 uptick in use of generative AI in assisting with the creation of personal statements.<sup>14,15</sup> Several  
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16 studies across multiple specialties have demonstrated that generative AI models can create high  
17  
18 quality personal statements that are often indistinguishable by expert faculty reviewers from  
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20 human written statements.<sup>16-20</sup> The ability to distinguish between the two types of content is  
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22 increasingly difficult as more detailed and advanced prompts with specific applicant details are  
23  
24 used to craft AI-generated statements.<sup>21</sup>  
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31 In addition to human review, numerous AI-detection tools have been created in response  
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33 to the growing use of generative AI in narrative writing as a way to flag possible AI-generated  
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35 content given the growing concern about authenticity and ethical use of AI. Use of AI-detection  
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37 tools on personal statements from non-pediatric residencies written before and after 2022 has  
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39 revealed a marked increase in the proportion flagged as AI-generated after 2022.<sup>14,22</sup>  
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43 In this study, we aimed to assess the percent of AI-generated text as determined by two  
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45 AI detection platforms in pediatric residency applicant personal statements and its frequency of  
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47 use between 2022 and 2024. We hypothesized that there would be an increase in AI detection  
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49 rates in pediatric residency personal statements over this timeframe as ChatGPT became  
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51 increasingly available with widespread use among the general population.  
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## 55 **Methods:**

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4 This was a retrospective, cross-sectional study at a single tertiary care academic center in  
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6 New England. Applicant rosters and materials including personal statements were downloaded  
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8 from the Electronic Residency Application System (ERAS) archives for those applying in 2022  
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10 through 2024. Applicant characteristics abstracted from the rosters included gender, self-identity,  
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12 medical school country of origin, and interview status. Applicants were excluded if no personal  
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14 statement was present in their application materials. Each personal statement was copied directly  
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16 from the application file and entered into two AI detection platforms: Quillbot (Quillbot.com)  
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18 and ZeroGPT (ZeroGPT.com). These platforms were chosen as they are both free, open-access  
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20 platforms and have been used in prior studies with good accuracy.<sup>21, 24</sup> We also used ChatGPT to  
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22 create an AI generated personal statement to test these platforms; both platforms reported 100%  
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24 of the text was AI generated. ChatGPT (Open AI Inc, San Francisco, CA) was released to the  
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26 public for commercial use on November 30, 2022, and hence we presumed that applicants would  
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28 not have had access to this technology for the 2022 application cycle. The Yale School of  
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30 Medicine Institutional Review Board deemed this study exempt from informed consent.  
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38 Quillbot indicates the percent of text that is likely AI, the percent that is AI generated, the  
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40 percent that is AI generated and AI refined, the percent that is human generated and AI refined,  
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42 and the percent that is human generated (Figure 1a). The percent of text that is likely AI is the  
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44 summative value of the AI generated and AI generated and AI refined percentages. ZeroGPT  
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46 provides only the percentage of text likely generated by AI (Figure 1b). The output from both AI  
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48 detectors was manually transcribed onto the data abstraction forms. To ensure consistent results,  
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50 a sampling of personal statements was re-run through both detectors.  
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55 Descriptive analyses were performed, including frequencies with percentages, means  
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57 with standard deviation (SD), and medians with interquartile ranges (IQR). Mean AI detection  
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between groups was compared using T-tests and ANOVA, whereas median values were analyzed with non-parametric tests. Medical school countries were categorized as United States (US) and Other. A small number of applicants identified as Alaskan-Americans or Native Americans; these applicants were grouped with applicants who self-identified as “Other” for our analyses. A general linear model was used to analyze the impact of AI detection on interview status while accounting for other applicant demographics and year. All analyses were performed in SPSS v 31.0 (IBM Corp.)

## Results:

There were a total of 3521 applicant files available for the study period. Personal statements were missing in 3 applications and were excluded from further analysis. Applicant characteristics and interview status for the sample and for each year of the study are presented in Table 1. Mean AI detection across all years was 11.7% (SD 26.3) in Quillbot at 13% (SD 20.5) in ZeroGPT. Mean AI detection in each platform and by applicant characteristics is presented in Table 2. The correlation between the total AI detected in each platform was .68 (95% CI .66, .7). There was a significant increase in mean AI detection in both platforms between 2022 and 2024 (Table 2). The proportion of applicants with any AI use also increased by year ( $p < .001$ , Figure 2). After adjusting for other applicant characteristics (eg. gender, self-identity, medical school country) and applicant year, there was no statistically significant difference in AI detection among applicants invited to interview and those who did not receive an invitation (mean difference 0.5, 95% CI -1.8, 2.8).

## Discussion:

Our study demonstrated an increase in AI generated content in pediatric residency personal statements over the timeframe of 2022-2024 across both AI detection platforms. We

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4 also found variability across AI detectors in how they classify text as AI- versus human-  
5 generated, as reflected in the moderate correlation between the two platforms. Both detectors  
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7 produced false positives, as labeled AI-generated content in personal statements from 2022 was  
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9 written before the public release of ChatGPT. These findings raise concern about the reliability  
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11 and accuracy of AI detectors, similar to what previous studies have shown.<sup>15, 23-25</sup>  
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17 While AI detection tools have become increasingly available, there is varying data on  
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19 their reliability and accuracy in detecting AI versus human generated content. Some studies  
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21 report numbers as high as 90% accuracy, while others are much lower (below 80% accuracy)  
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23 across multiple detectors.<sup>23-26</sup> In addition, AI detectors often misclassify non-native English  
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25 writing as AI-generated, creating the potential for bias.<sup>27</sup> These challenges highlight the need to  
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27 better understand how AI detection tools perform in contexts that have not yet been studied  
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29 extensively, including pediatric residency applications.<sup>14,27</sup>  
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35 When reviewing AI-generated content use across demographics, we found that applicants  
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37 who attended non-U.S. medical schools had higher AI detection rates than those who attended  
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39 U.S. medical schools, as well as differences in AI-detection rates across self-identified  
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41 race/ethnicity categories. Notably, those applicants who self-identified as Middle Eastern or  
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43 “Other” had higher means of AI-detection, particularly in comparison to those applicants who  
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45 self-identified as White. This raises concern about AI detectors misclassifying written text by  
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47 non-native English speakers and supports the notion that detecting AI-generated content remains  
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49 difficult and that results from an AI-detector must be interpreted with caution.<sup>27</sup> It is also  
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51 possible that non-native English speakers use AI more often to help optimize language, style, or  
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53 grammar in their personal statements.  
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4 Reality suggests that generative AI is here to stay, and its integration into everyday life  
5 and professional practice will continue to expand. When third and fourth year medical students  
6 were surveyed about their intentions to use AI in the creation of their personal statements for  
7 residency applications, more than half of the responders in the survey reported having already  
8 used or were planning to use generative AI for assistance when writing their personal  
9 statements.<sup>28</sup> Our study data support these findings with some AI detection present in a large  
10 proportion of residency applicants' personal statements. In contrast, when plastic surgery faculty  
11 were surveyed about the use of generative AI in residency personal statements, most believed it  
12 to be unethical, with a minority supporting its use for editing personal statements or  
13 brainstorming ideas. A very small minority supported its use in all aspects of creation and  
14 writing of the personal statement.<sup>29</sup>  
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32 Having insight into which applicants used AI in creating their personal statement may  
33 also introduce another level of bias to the residency selection process. Program directors have  
34 raised concerns about an applicant's work ethic if they knew that they used generative AI in the  
35 creation of their personal statement, which could negatively impact their candidacy.<sup>30,31</sup> Without  
36 having reliable AI detection, it may be dangerous to prematurely make a potentially false  
37 assumption. We did not find a significant difference in interview status between applicants who  
38 appeared to have AI-generated personal statements compared to those that were human written.  
39 This may be related to our pediatric residency's process of holistic review for applicants and  
40 which factors play into offering an interview, in addition to previously discussed limitations  
41 regarding the potential misclassification of AI versus human generated text by an AI detector as  
42 well as the ability for AI models to generate high quality personal statements. Overall, other  
43 studies to date have reported mixed findings: some show higher interview rates for ChatGPT-  
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generated statements (when reviewers are blinded to AI vs. human generated content), while others demonstrate that statements perceived as human-written by expert reviewers are more likely to result in interview invitations.<sup>16,32</sup>

There are several limitations to our study. First, our study represents retrospective data from a single academic center and medium-sized residency program in New England, and this may limit generalizability, as applicant pools and selection criteria for pediatric residency may vary across geographic regions and program sizes.

Second, given the known variability in the accuracy of the two AI detectors used in this study, our results may include both false positives and false negatives in terms of AI versus human-generated content. Given that this was a retrospective study of ERAS applications, we could not confirm whether or not applicants used AI in the creation of their personal statements, making AI detector output an imperfect proxy for AI use. In addition, we also only used two AI detectors in our analysis, both of which were free versions and use different algorithms in their models for detecting AI generated content. Also, although ERAS applications indicate language fluency, this field changed over the three years of our study collection, precluding a more nuanced analysis of preferred language and AI use. Lastly, AI familiarity, use, and sophistication rapidly grew between 2022 and 2024; if AI detector technology did not evolve at a similar pace, then their ability to distinguish AI from human-generated text may have changed over time.

Given the current limitations of AI detectors, residency PDs should approach AI using detection cautiously. If residency PDs choose to use commercially available AI detectors as part of their workflow in screening personal statements, they must be aware of the potential bias they create without knowing with certainty how an applicant used (or did not use) AI in the creation of their personal statement. Residency applicants' use of AI in their personal statements makes it

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4 increasingly difficult to differentiate between candidates during the screening process and poses  
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6 significant challenges for consistent enforcement. It is possible that AI detector technology will  
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8 continue to improve with ongoing training and refinement similar to AI-generated technology,  
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10 and this will warrant ongoing surveillance. Future work should prioritize establishing standards  
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12 for ethical use around AI in residency applications as this is a technology that will continue to  
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14 advance and will be widely used by the general public.  
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## 21 **Conclusion:**

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23 Generative AI use in pediatric residency applications is increasing and will continue to  
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25 evolve as large language models advance and become integrated into daily life. Rather than  
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27 discouraging the use of generative AI outright, educators should work towards establishing best  
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29 practices that support applicants using AI responsibly, including disclosure of use. This  
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31 underscores the importance of creating clear guidance on its ethical and transparent use. Given  
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33 that current AI detectors remain unreliable and risk introducing bias into the residency selection  
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35 process, PDs should use them cautiously.  
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Table 1. Applicant demographics and characteristics

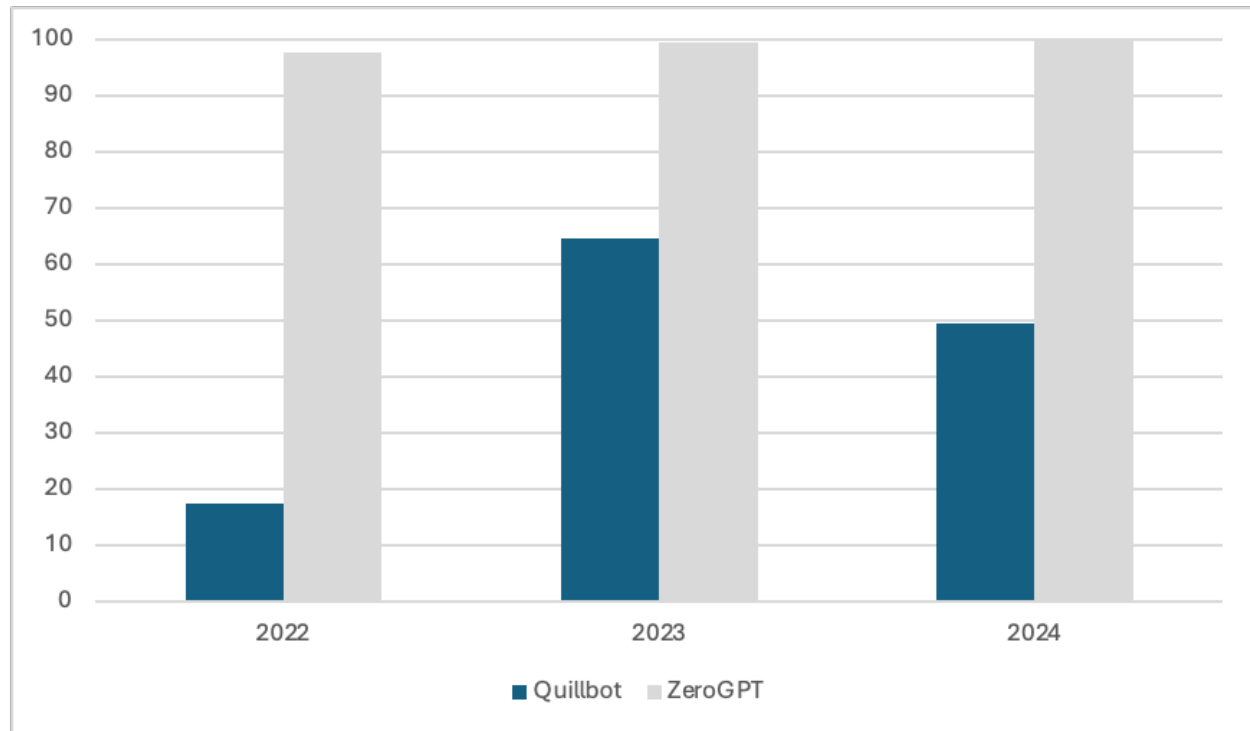
| Characteristic            | Total<br>N=3521 | 2022<br>N=1225 | 2023<br>N=1139 | 2024<br>N=1157 |
|---------------------------|-----------------|----------------|----------------|----------------|
| Gender                    |                 |                |                |                |
| Male                      | 780 (22.2%)     | 281 (22.9%)    | 235 (20.6%)    | 264 (22.8%)    |
| Female                    | 2728 (77.8%)    | 939 (76.7%)    | 898 (78.8%)    | 891 (77%)      |
| Missing                   |                 | 5 (0.4%)       | 6 (0.5%)       | 2 (0.2%)       |
| Self-Identity             |                 |                |                |                |
| Asian                     | 1067 (30.3%)    | 364 (29.7%)    | 349 (30.6%)    | 354 (30.6%)    |
| Black                     | 328 (9.3%)      | 108 (8.8%)     | 107 (9.4%)     | 113 (9.8%)     |
| Hispanic                  | 486 (13.8%)     | 170 (13.9%)    | 160 (14%)      | 156 (13.5%)    |
| Middle Eastern            | 135 (3.8%)      | 19 (1.6%)      | 22 (1.9%)      | 94 (8.1%)      |
| White                     | 1327 (37.7%)    | 507 (41.4%)    | 442 (38.8%)    | 378 (32.7%)    |
| Other                     | 75 (2.1%)       | 28 (2.8%)      | 21 (1.8%)      | 26 (2.2%)      |
| Missing                   | 103 (2.9%)      | 29 (2.4%)      | 38 (3.3%)      | 36 (3.1%)      |
| Medical School<br>Country |                 |                |                |                |
| US                        | 2198 (62.4%)    | 789 (64.4%)    | 722 (63.4%)    | 687 (59.4%)    |
| Other                     | 1323 (37.6%)    | 436 (35.6%)    | 417 (36.6%)    | 470 (40.6%)    |
| Interviews                | 671 (19.1%)     | 254 (20.7%)    | 249 (21.9%)    | 168 (14.5%)    |

Table 2. Mean AI detection in each platform by applicant characteristics

\*p&lt;.05 \*\*p&lt;.001

| Mean AI Detection (SD) | Quillbot     |              |                             |                              |               | ZeroGPT     |
|------------------------|--------------|--------------|-----------------------------|------------------------------|---------------|-------------|
|                        | % Likely AI  | AI Generated | AI Generated and AI refined | Human Written and AI refined | Human Written |             |
| All Applicants         | 19.59 (28.3) | 16.31 (26)   | 3.12 (7)                    | 1.21 (3.3)                   | 79.2 (28)     | 50.6 (28.1) |
| Gender                 |              |              |                             |                              |               | **          |
| Male                   | 18.2(28.1)   | 15.1(25.5)   | 3(7.1)                      | 1.3(3.5)                     | 80.4(27.9)    | 46.6(28.4)  |
| Female                 | 20.1(28.4)   | 16.7(26.1)   | 3.2(6.9)                    | 1.2(3.2)                     | 78.8(28.1)    | 51.8(27.9)  |
| Self-Identity          | **           | **           | **                          |                              | **            | **          |
| Asian                  | 23.4(29.5)   | 19.4(27.3)   | 3.7(7.3)                    | 1.1(3.2)                     | 75.5(29.2)    | 51.7(28.1)  |
| Black                  | 23.4(32.7)   | 20(30.3)     | 3.1(7.3)                    | 1.1(3.1)                     | 75.5(32.4)    | 54.2(30.2)  |
| Hispanic               | 20(27.1)     | 16.4(24.4)   | 3.6(7.4)                    | 1.2(3.3)                     | 78.8(26.9)    | 53(27.9)    |
| Middle Eastern         | 33.5(33.9)   | 28.8(32.1)   | 4.7(8.7)                    | .8(2.7)                      | 65.7(33.7)    | 60.7(30.6)  |
| White                  | 13.5(23.8)   | 11(21.5)     | 2.4(6.1)                    | 1.4(3.5)                     | 85.1(23.6)    | 46.6(26.5)  |
| Other                  | 23.8(32.6)   | 21.5(31.1)   | 2.4(4.5)                    | 1(3.8)                       | 75.2(32.3)    | 50.8(28.5)  |
| Medical School         | **           | **           | **                          |                              | **            | **          |
| Country                |              |              |                             |                              |               |             |
| US                     | 15(24.6)     | 12.3(22.3)   | 2.5(6.1)                    | 1.2(3.2)                     | 83.8(24.4)    | 47.8(26.3)  |
| Other                  | 27.3(32.1)   | 22.9(29.9)   | 4.1(8.1)                    | 1.3(3.5)                     | 71.5(31.6)    | 55.2(30.1)  |
| Interviewed            | **           | **           | *                           |                              | **            | **          |
| Yes                    | 14.2(22.8)   | 11.5(20.5)   | 2.5(6)                      | 1.1(2.9)                     | 84.7(22.6)    | 46.8(25.2)  |
| No                     | 20.9(29.3)   | 17.5(27)     | 3.3(7.2)                    | 1.2(3.4)                     | 77.9(29)      | 51.5(28.6)  |
| Year                   | **           | **           | **                          |                              | **            | **          |
| 2022                   | 4.9(12.9)    | 3.4(10)      | 1.4(4.9)                    | 1.1(3.2)                     | 93.9(13.1)    | 35.9(21.1)  |
| 2023                   | 22.6(24.9)   | 17.7(22.6)   | 4.9(8.1)                    | 1.2(3.2)                     | 76.1(24.6)    | 51.5(26.2)  |
| 2024                   | 32.1(35.4)   | 28.6(33.4)   | 3.2(7.3)                    | 1.3(3.5)                     | 66.6(34.8)    | 65.3(28.3)  |

Figure 2. Proportion of applicants with any amount of AI detection in Quillbot and ZeroGPT by year.



Dear Dr. Sullivan,

Please accept our contribution to the literature entitled: Use of Generative Artificial Intelligence in Pediatric Residency Personal Statements. We believe it will make three important contributions to the current field of artificial intelligence use in medical education:

1. To our knowledge, this study is relatively novel in investigating the use of generative artificial intelligence (AI) by pediatric residency applicants.
2. This manuscript supports the rising use of artificial intelligence use by pediatric residency applicants and the challenge in detection of human-generated versus AI-generated content in their personal statements.
3. This manuscript will provide further guidance for the considering the establishment of best practices for responsible artificial intelligence use by pediatric residency applicants, including disclosure.

All authors participated in the conception, design, execution, and writing of the manuscript. The authors have no personal financial disclosures to report. This study was approved by the Yale School of Medicine IRB, which waived the need for informed consent.

We wish to note that this manuscript is being submitted alongside a companion paper entitled "Growing Use of Artificial Intelligence in Pediatric Fellowship Personal Statements." The two manuscripts report complementary findings yet address distinct research questions and have been prepared as separate submissions for focused audiences.

Thank you for your time and consideration.

Sincerely,



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